$s(x) \longmapsto x \cdot s(x)$ (mod)

i.e. [0x3b]

$\mathbb{F}_2^{32}$

note: eax s

$s(x) \in \mathbb{F}_2[x]/(p(x))$

Galois LFSR

0xc0000401

$\mathbb{F}_2[x] \ni p(x) = x^{32} + x^{31} + x^{30} + x^{10} + 1$

i.e. multiplies by $x \in \mathbb{F}_2[x]/(p(x))$

{multiply P(x) by x ∀ sign bit of} → eax → edx:eax

```
cdq
shl     eax, 1
and     edx, 0xc0000401
xor     eax, edx
```

edx₀ < 0

edx = 0xc0000401

eax₀ ≥ 0

eax = 2 → eax₀ << 1 = eax

$16^7 \cdot c + 16^2 \cdot 4 + 1$

edx = 0

IF (eax₀ ≥ 0): → eax ≥ 0, doubling

(eax₀ << 1)

⊕ 0

(eax₀ << 1)

ELSE (eax < 0): → eax < 0,

(eax₀ << 1)

⊕ 0xc000 0401

(eax << 1)

⊕ 0xc000 0401

outputs are always non-negative

$(16^7 \cdot 12) + (16^2 \cdot 4) + 1$

0101

$P(x) = x^{32} + (c_{31}x^{31} + c_{30}x^{30} + c_{10}x^{10} + c_0(x^0))$

if neg, make positive

& xor to set

31-bit, 30-bit, 10-bit, 0-bit

???

[0x3b] meaning.

$p(x) = x^{31} + x^{30} + x^{10} + 1$??